AI-Driven Assessment Development

Automating Item Generation, Fairness, & Validation

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Scan to access the pipeline

github.com/tariq-shaban/...

SIOP 2026 • Session 970 • April 30, 9:00–10:20 AM • Room R7

Scale development doesn't scale

3–6

months

per assessment

$50K+

typical project cost

500+

items

reviewed manually

Most of that time is spent on generative and screening stages — exactly where LLMs and NLP can reduce effort without compromising rigor.

Guenole et al. (2024) • Lee et al. (2025) • Suárez-Álvarez et al. (2026)

What is Automatic Item Generation?

Dr. Philseok Lee

AIG uses algorithms to automatically generate test items

Problems with Traditional Item Writing

- Time-consuming
- Expensive
- Limited item pool
- Human bias

AIG Advantages

- Generate hundreds/thousands of items quickly
- Improve test security (less item exposure)
- Enable adaptive testing

Template-Based AIG

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TEMPLATE-BASED AIG

Historically used in educational aptitude testing

"A 68-kg, 50-year-old security guard was **[Location]** during an altercation with another man. He is immediately brought to the hospital where his vitals are as follows: **[Blood Pressure]** and **[Heart Rate]**. He has **[Air Entry]**. His neck veins are **[Neck Veins]** and his trachea is **[Trachea]**.

What is the most likely diagnosis?"

Limits: struggles with sophisticated semantic meaning, subtle nuance, and rich situational context (Hommel et al., 2022).

LLM-BASED AIG

What this pipeline uses

- Generates from construct definitions
- Flexible and context-rich output
- Particularly effective for noncognitive AIG — personality, values, SJTs
- Enables construct-anchored behavioral specificity

Evidence base: Götz et al. (2024); Hernandez & Nie (2023); Hommel et al. (2022); Lee et al. (2023, 2025); von Davier (2018)

Pipeline architecture

Dr. Philseok Lee & Tariq Shaban

Five phases • one notebook • full audit trail

1
Behavioral Domains
`01_behavioral_domains.xlsx`

→

2
Item Generation
`02_generated_items.xlsx`

→

3
Readability & Bias
`03_readability_bias.xlsx`

→

4
Content Validity
`04_content_validity.xlsx`

→

5
Pseudo-Factor Analysis
`05_pseudo_factor_analysis.xlsx`

Human-in-the-loop at every phase

Every phase writes `human_review_pass` and `human_comments`. Every downstream phase reads them. Experts have full veto power without touching code.

"Responsible use requires ongoing human oversight." — Suárez-Álvarez et al. (2026)

Three moving parts

Tariq Shaban

What you edit vs. what runs

scales.json

what you're measuring

Scale names, construct definitions, high / low anchors, discriminant descriptions

```
{
  "scale_name":
    "Conscientiousness",
  "construct_definition":
    "Tendency to be
     organized, diligent...",
  "high_anchor":
    "Works in an
     organized way...",
  "discriminant_description":
    "Self-regulation, not
     moral conduct..."
}
```

- *Edit to measure something new*

pipeline_settings.json

how the pipeline behaves

Model choices, thresholds, pass rules, item counts, encoding strategy

```
{
  "transformer_models": [
    "all-MiniLM-L6-v2"
  ],
  "pass_rules": {
    "Rule_1_Strong_Single_
        Factor": {
      "threshold": 0.40
    }
  },
  "items_per_keying": {
    "positive": 10,
    "negative": 5
  }
}
```

- *Edit to tune behavior or change LLMs*

majdal_pipeline.ipynb

the engine

Five phases as code cells — reads the JSONs, writes the Excel outputs

```
# Phase 1
generate_domains()

# Phase 2
generate_items()
flag_duplicates()

# Phase 3
readability_bias()

# Phase 4
content_validity()

# Phase 5
pseudo_factor_analysis()
```

- *Usually unchanged — JSONs drive everything*

Change ***what*** and ***how*** in the JSONs. Run the same notebook unchanged.

Phase 1

Behavioral Domains

From construct definitions to validated behavioral examples

Tariq Shaban

Phase 1: behavioral domains

scales.json — Conscientiousness

```
{
  "scale_name": "Conscientiousness",

  "construct_definition":
    "The tendency to be organized,
     diligent, and goal-oriented. The
     capacity to regulate one's own
     behavior to achieve desired
     outcomes.",

  "high_anchor":
    "Works in an organized way;
     meets commitments.",

  "low_anchor":
    "Approaches tasks
     unsystematically; misses
     deadlines.",

  "discriminant_description":
    "Conscientiousness concerns
     self-regulation, not moral
     conduct (see Honesty-Humility)."
}
```

1. **LLM generates behavioral domains**
   Observable, recurring patterns — not one-off incidents
2. **Occurrence likelihood assigned**
   high / moderate / low — construct breadth, not difficulty
3. **Second LLM validates each domain**
   Construct, face, and discriminant validity
4. **Failed domains flagged for human review**
   Override any decision in the Excel output

GOOD "Following workplace rules and procedures" **BAD** *"An employee updates their project tracker daily" (an incident, not a domain)*

LIVE DEMO

Phase 1

Code walkthrough + execution

Phase 2

Item Generation

From behavioral domains to survey items

Dr. Philseok Lee

Phase 2: dominance-item framework

1 Dominance framework

Standard Likert — higher trait = stronger endorsement. The format used by every operational personality inventory.

Clark & Watson 1995; DeVellis 2017

2 Keying direction

Positive and negative items generated separately, default 2:1 ratio. Controls acquiescence bias.

3 Domain specificity

Every item rooted in its source behavioral domain. Generic items rejected by prompt rules.

4 Sentence-BERT duplicate detection

Semantic similarity — catches paraphrase duplicates lexical TF-IDF misses.

swap recommended by Dr. Lee

Worked example — Conscientiousness

Domain: "Following workplace rules and procedures"

POSITIVE "I consistently follow the rules at work"

NEGATIVE "I often look for ways around workplace rules"

NOT A REVERSAL "I sometimes forget workplace rules"

Low-intensity positive, not a true reversal. A negative item must be something a HIGH scorer clearly DISAGREES with.

Item writing rules — enforced by the prompt

- ✓ "I" statement in present tense
- ✓ Maximum 9 words (including "I")
- ✓ One idea only — no "and" / "but" / "or"
- ✓ Plain everyday vocabulary
- ✓ No abstract nouns ("efficiency", "integrity")
- ✓ Observable behavior — not beliefs or values
- ✓ Domain-specific, not construct-generic
- ✓ Social desirability neutralized

Dominance framework references Clark & Watson (1995); DeVellis (2017)

LIVE DEMO

Phase 2

Code walkthrough + execution

Phase 3

Readability & Bias

Automated screening with human review of edge cases

Tariq Shaban

Phase 3: readability & bias

READABILITY SCREENING

6 metrics per item: Flesch-Kincaid, Flesch Reading Ease, Gunning Fog, Coleman-Liau, SMOG, ARI

- Items above reading-level cap (default Grade 12) simplified automatically by LLM
- Up to max_simplification_attempts retries per item
- Original wording preserved in original_item_text for audit
- Negatively keyed items retain reversal language during simplification

LLM BIAS REVIEW

- 3 independent LLM reviewers rate every item on 6 bias dimensions
- Flag threshold: mean < 3.5 on any dimension (asymmetric rubric)
- Replaces regex matching with semantic evaluation
- Items not excluded automatically — flagged with concern notes for human review
- *Dimensions: gender, ethnicity / race, age, religion / culture, socioeconomic, sexual orientation / disability*

Zieky (2013); Lee, Son & Jia (2025) LM-AIG framework

AERA / APA / NCME Standards (2014) — addressed at scale

Standard 3.0 Item development procedures documented, construct-based

Standard 1.9 Content validity evidence from multiple qualified reviewers

+ SIOP AI Assessment Guidelines (2023)

Standard 3.2 Items minimize construct-irrelevant variance (reading, bias)

Standard 7.1 Fairness addressed at design stage, not only post-hoc

Automated screening applies these at scale; human review ensures expert judgment is the final arbiter.

LIVE DEMO

Phase 3

Code walkthrough + execution

Phase 4

Content Validity

Five independent AI reviewers rate every item

PHASE 4

Dr. Philseok Lee

Phase 4: correspondence & distinctiveness

SME1	SME2	SME3	SME4	SME5
Claude Haiku	**GPT-4o**	**Llama 3.3**	**Grok 3 Mini**	**DeepSeek**
Anthropic	*OpenAI*	*Meta*	*xAI*	*DeepSeek*

Each SME rates the item against ALL scales on a 1–7 Likert scale

c-value

Correspondence

How well does the item measure what it's supposed to measure?

`mean target rating / 7`

Threshold: **≥ 0.88**

d-value

Distinctiveness

How clearly does the item separate from constructs it shouldn't measure?

`(target - mean other) / (a-1)`

Threshold: **≥ 0.35**

Pass rule: **c-value ≥ 0.88** AND **d-value ≥ 0.35** — *both must be met*

Hinkin & Tracey (1999) • Colquitt et al. (2019) • Lee, Son & Jia (2025) LM-AIG

LIVE DEMO

Phase 4

Code walkthrough (pre-computed output loaded)

Phase 5

Pseudo-Factor Analysis

Embedding-based factor analysis before any human data

The substitutability assumption

Embedding vector ≈ aggregate semantic encoding

Response vector ≈ aggregate empirical encoding

Semantic similarity ≈ item co-occurrence in human data

"Both vector formats represent aggregated encodings of personality statements... heuristic motivation to investigate alignment." — Guenole et al. (2024)

PHASE 5

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Phase 5: PFA mechanics + discriminant validity

1
Encode
Sentence-BERT per item

→

2
Similarity
Cosine matrix (averaged if >1 model)

→

3
Factor analysis
PCA on similarity matrix

→

4
6 pass rules
HIGH_PASS or STANDARD_PASS

```
transformer_models: ["all-MiniLM-L6-v2"]     # add "all-mpnet-base-v2" for multi-model averaging (Guenole et al., 2024)

pass_rules: Rule_1_Strong_Single_Factor ≥ 0.40  •  Rule_3_Adequate_Communality ≥ 0.10  •  Rule_4_Clean_Factor_Structure < 0.35
```

DAAL factor identity

Dominant Average Absolute Loading (Guenole et al., 2024)

- Joint multi-scale factor analysis
- Each factor labeled by the scale with highest mean |loading|
- Scale's DAAL identity "confirmed" when its items load strongest on its own factor

Discriminant validity

Target-to-max-cross-loading ratio

- Target loading ÷ highest cross-loading on another scale
- Flag when ratio < 1.5 OR any cross-loading > 0.45
- Catches construct contamination that pass-rules alone would miss

PHASE 5

Dr. Philseok Lee

Converging evidence for PFA

1	**Guenole et al. (2024)**	PFA recovers NEO & HEXACO structures; multi-transformer averaging outperforms single models
2	**Milano et al. (2025)**	LLM embeddings predict a-priori factorial structure across 4 personality tests
3	**Hommel & Arslan (2025)**	Fine-tuned SurveyBot3000: item r = .59, scale r = .84 with human responses
4	**Varrasi et al. (2026)**	PFA extends to clinical/health measures; excellent English recovery, variable cross-language
5	**Suárez-Álvarez et al. (2026)**	First independent practical guide for AI in test construction; confirms PFA utility
6	**Jia & Lee (under review)**	Full psychometrics development profile including fine-tuning and synthetic data generation

LIVE DEMO

Phase 5

Code walkthrough + execution

LIVE DEMO

Output walkthrough

"Can the pipeline be tricked?"

`Review_Pool → Recommendations → Quality_Flags → PFA_Item_Results`

The setup: 5 Honesty-Humility items injected into Phase 4 with **forged c-values of 0.95**, each mislabeled as a different Big Five scale.
Question: does the pipeline catch them at Phase 5?

PHASE 5

Limitations & future directions

Current limitations

- PFA screens — it does not replace empirical data
- Substitutability varies across constructs
- Cross-linguistic performance is variable
- Model choice affects quality significantly
- No IRT integration yet

Future directions

- Fine-tuned transformers for psychometric embeddings
- Cross-cultural and multilingual validation
- IRT model integration for difficulty estimation
- Automated forced-choice block construction
- Live updating from pilot data → re-run PFA

"We do not propose to fully automatize scale construction... at the end of the day, empirical research on real data is always required."

— Guenole, D'Urso, Samo, & Sun (2024)

Beyond scales: other applications

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BEHAVIORAL COMPETENCIES

- Same pipeline architecture adapts directly
- Semantic similarity validates that competency definitions are distinct
- LLM generates behavioral indicators at multiple proficiency levels
- Embedding analysis flags overlapping competencies before field testing

SITUATIONAL JUDGMENT TESTS

- Multi-agent: scenario → options → validation
- Readability and bias screening transfers directly
- Semantic similarity evaluates whether response options discriminate between constructs
- PFA principles apply to response scoring keys

The next two slides show what's possible when the same architecture extends to multimedia scenario production.

AI-Generated SJT · Cinematic style

AI-Generated SJT · Stylized 3D cartoon rendered

Three takeaways

Dr. Philseok Lee & Tariq Shaban

1

The tooling exists today

Open-source, working pipeline — scan and download

2

The science is solid and growing

Converging evidence across independent research groups

3

Humans remain in control

Every decision overridable, every output auditable

Let's talk: ethical considerations

- Who is responsible when an automated pipeline produces a biased item?
- What does "informed consent" mean when test-takers face AI-generated content?
- How do we audit a pipeline that's faster than human review can keep up with?

Thank you — Questions?

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Automated-Psychometric-Scale-Development-Pipeline

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Other SIOP 2026 sessions you might find related:

Session 578 Training AI Skills in I-O Graduate Training (Apr 30, 8:00 AM)
Session 777 Forced-Choice Testing: New Methodologies (May 1, 2:00 PM)
Session 961 AI, Fairness, and the Future of Personality Testing (May 2, 3:00 PM)
Session 41 Novel Applications of LLM in Org Measurement (Apr 30, 5:00 PM)
Session 848 Protecting Assessment Integrity in AI Era (May 2, 8:00 AM)

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